

MICROECONOMICS II

About the course

Topic 1 - Technology

Julie Chytilová
julie.chytilova@fsv.cuni.cz

LECTURERS

Julie Chytilová

- ▶ Office hours: Friday 10:30-11:45 am (please email me first)
- ▶ Office 410
- ▶ Email: julie.chytilova@fsv.cuni.cz

Matěj Bajgar

- ▶ Office hours by appointment
- ▶ Office 406
- ▶ Email: matej.bajgar@fsv.cuni.cz

TEACHING ASSISTANTS

Seminars

- ▶ Diana Kmet'ková: diana.kmetkova@fsv.cuni.cz
- ▶ Daniel Sůva: daniel.suva@fsv.cuni.cz

Home assignment

- ▶ Diana Kmet'ková: diana.kmetkova@fsv.cuni.cz

Final exam

- ▶ Iñaki Alberto Veruete Villegas: inaki.veruete@czp.cuni.cz

CLASSES

Lectures

- ▶ Thursday 9:30-10:50am in room 109 plus every other week on Thursday 11:00am-12:20pm in room 109

Joint seminars for all students

- ▶ Every other week on Thursday 11:00am-12:20pm in room 109

Regular seminars where the students are divided into three groups.

- ▶ Monday 8:00-9:20am in room 206
- ▶ Thursday 8:00-9:20am in room 314

It is recommended that the students attend all lectures (blue color in the table), all joint seminars (green) and regular seminars (light or dark yellow).

CLASSES

Please see the syllabus for a visible version of this table.

Day	Date	Time	Room	Instructor	
Thursday	2-Oct	9:30-10:50am	109	Lecture 1	Juke Chyčková
Thursday	2-Oct	11:00am-12:20pm	109	Lecture 2	Juke Chyčková
Monday	6-Oct	8:00-9:20am	206	Seminar 1	Diana Knerelková
Thursday	9-Oct	8:00-9:20am	314	Seminar 1	Diana Knerelková
Thursday	9-Oct	9:30-10:50am	109	Lecture 3	Juke Chyčková
Thursday	9-Oct	11:00am-12:20pm	109	Seminar 2	Daniel Siva
Monday	13-Oct	8:00-9:20am	206	Seminar 3	Diana Knerelková
Thursday	16-Oct	8:00-9:20am	314	Seminar 3	Diana Knerelková
Thursday	16-Oct	9:30-10:50am	109	Lecture 4	Juke Chyčková
Thursday	16-Oct	11:00am-12:20pm	109	Lecture 5	Juke Chyčková
Monday	20-Oct	8:00-9:20am	206	Seminar 4	Diana Knerelková
Thursday	23-Oct	8:00-9:20am	314	Seminar 4	Diana Knerelková
Thursday	23-Oct	9:30-10:50am	109	Lecture 6	Juke Chyčková
Thursday	23-Oct	11:00am-12:20pm	109	Seminar 5	Daniel Siva
Monday	27-Oct	8:00-9:20am	206	Dean's day	
Thursday	30-Oct	8:00-9:20am	314	no seminar	
Thursday	30-Oct	9:30-10:50am	109	no lecture	
Thursday	30-Oct	11:00am-12:20pm	109	no lecture/seminar	
Monday	3-Nov	8:00-9:20am	206	Seminar 6	Diana Knerelková
Thursday	6-Nov	8:00-9:20am	314	Seminar 6	Diana Knerelková
Thursday	6-Nov	9:30-10:50am	109	Lecture 7	Juke Chyčková
Thursday	6-Nov	11:00am-12:20pm	109	Lecture 8	Matěj Bažgar
Monday	10-Nov	8:00-9:20am	206	Seminar 7	Diana Knerelková
Thursday	13-Nov	8:00-9:20am	314	Seminar 7	Diana Knerelková
Thursday	13-Nov	9:30-10:50am	109	midterm	
Thursday	13-Nov	11:00am-12:20pm	109	seminar midterm	Daniel Siva
Monday	17-Nov	8:00-9:20am	206	Public holiday	
Thursday	20-Nov	8:00-9:20am	314	Seminar 8	Diana Knerelková
Thursday	20-Nov	9:30-10:50am	109	Lecture 9	Matěj Bažgar
Thursday	20-Nov	11:00am-12:20pm	109	Lecture 10	Matěj Bažgar
Monday	24-Nov	8:00-9:20am	206	Seminar 9	Daniel Siva
Thursday	27-Nov	8:00-9:20am	314	Seminar 9	Daniel Siva
Thursday	27-Nov	9:30-10:50am	109	Lecture 11	Matěj Bažgar
Thursday	27-Nov	11:00am-12:20pm	109	Seminar 10	Daniel Siva
Monday	1-Dec	8:00-9:20am	206	Seminar 11	Daniel Siva
Thursday	4-Dec	8:00-9:20am	314	Seminar 11	Daniel Siva
Thursday	4-Dec	9:30-10:50am	109	Lecture 12	Matěj Bažgar
Thursday	4-Dec	11:00am-12:20pm	109	Lecture 13	Matěj Bažgar
Monday	8-Dec	8:00-9:20am	206	Seminar 12	Daniel Siva
Thursday	11-Dec	8:00-9:20am	314	Seminar 12	Daniel Siva
Thursday	11-Dec	9:30-10:50am	109	Lecture 14	Matěj Bažgar
Thursday	11-Dec	11:00am-12:20pm	109	Seminar 13	Daniel Siva
Monday	15-Dec	8:00-9:20am	206	Seminar 14	Daniel Siva
Thursday	18-Dec	8:00-9:20am	314	Seminar 14	Daniel Siva
Thursday	18-Dec	9:30-10:50am	109	Seminar recap	Daniel Siva
Thursday	18-Dec	10:40am-12:20pm	109	Final exam 1	
Thursday	15-Jan	9:30-11:00am	109	Final exam 2	
Thursday	22-Jan	9:30-11:00am	109	Final exam 3	
Thursday	5-Feb	9:30-11:00am	109	Final exam 4	
Thursday	11-Dec	9:30-10:50am	109	Lecture 14	Matěj Bažgar
Thursday	11-Dec	11:00am-12:20pm	109	Seminar 13	Daniel Siva
Friday	12-Dec	8:00-9:20am	206	Seminar 14	Daniel Siva
Monday	15-Dec	8:00-9:20am	206	Seminar 14	Daniel Siva
Monday	18-Dec	8:00-9:20am	314	Seminar 14	Daniel Siva
Thursday	18-Dec	9:30-10:50am	109	Seminar recap	Daniel Siva
Thursday	18-Dec	10:40am-12:20pm	109	Final exam 1	
Thursday	15-Jan	9:30-11:00am	109	Final exam 2	
Thursday	22-Jan	9:30-11:00am	109	Final exam 3	
Thursday	5-Feb	9:30-11:00am	109	Final exam 4	

■ Lectures
■ Super-seminars
■ Monday seminars
■ Thursday seminars

TOPICS

Theory of producer behavior and market structure

- ▶ Technology
- ▶ Profit maximization
- ▶ Cost minimization
- ▶ Firm supply
- ▶ Industry supply
- ▶ Monopoly, price discrimination
- ▶ Oligopoly



Why do the keypad buttons on drive-up cash machines have Braille dots?



Why is milk sold in rectangular containers and soft drinks in round ones?



Why do budget airlines charge for onboard meals (free on luxury airlines), while luxury hotels charge for Internet access (free at budget hotels)?

LITERATURE

Hal R. Varian, Intermediate Microeconomics: A Modern Approach, W.W. Norton and Company, 8th edition, 2010.

- ▶ Many copies are available in the IES library.
- ▶ Any extensions presented at the lectures and not available in the Varian textbook can be downloaded from SIS.

Other useful textbooks:

- ▶ T. Nechyba, Microeconomics: An Intuitive Approach with Calculus, South-Western, Cengage Learning
- ▶ W. Nicholson/C. Snyder, Microeconomic Theory: Basis Principles and Extensions, South- Western
- ▶ Schotter, Microeconomics: A Modern Approach, South-Western 2009

GRADING

- ▶ Home assignment (10 points)
- ▶ Mid-term exam (30 points)
- ▶ Final exam (60 points)

Final grade

- ▶ A: 90.1-100
- ▶ B: 80.1-90
- ▶ C: 70.1-80
- ▶ D: 60.1-70
- ▶ E: 50.1-60
- ▶ F: 0-50

SCHEDULE

Home assignment

- ▶ Released on November 20, submission deadline on November 30.

Midterm exam: November 13th, 2025, 9:30-10:50am, in room 109.

There will be one midterm exam only. If you cannot attend the midterm exam you will be requested to provide confirmation that your health condition prevented you from attending it.

SCHEDULE

Final exam

- ▶ December 18, 2025, 10:40am-12:20pm
- ▶ January 15, 2026, 9:30-11:10am
- ▶ January 22, 2026, 9:30-11:10am
- ▶ February 5, 2026, 9:30-11:10am

Topic 1 - Technology

INPUTS AND OUTPUTS

Constraints imposed by customers, competitors, **nature**.

Technology: transformation of inputs in outputs

Factors of production: Land, labor, raw materials, (physical) capital

Flow units

Assumptions

- ▶ Costless not to use all inputs
- ▶ Perfect divisibility of inputs and outputs
- ▶ Perfect information
- ▶ Non-existence of externalities
- ▶ Homogeneity of inputs and outputs
- ▶ More assumptions later on

TECHNOLOGICAL CONSTRAINTS

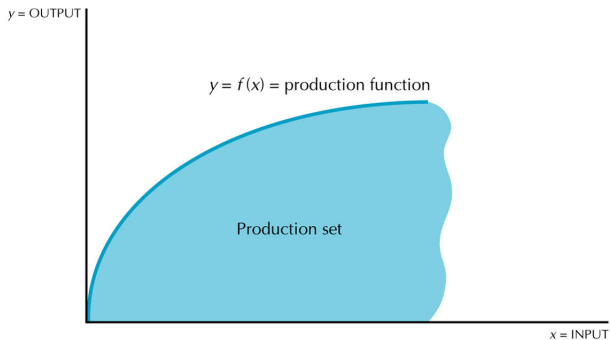


FIGURE 18.1 A production set

Intermediate Microeconomics, 8th Edition
Copyright © 2010 W. W. Norton & Company

TECHNOLOGICAL CONSTRAINTS

Production set

- ▶ All feasible combinations of inputs and outputs
- ▶ $Y = \{(y, x_1, \dots, x_n) | 0 \leq y \leq f(x_1, \dots, x_n); x_i \geq 0 \forall i\}$

Production function

- ▶ Upper boundary of the production set
- ▶ Production function f assigns to each bundle of inputs $(x_1, \dots, x_n)$ the maximum possible output which can be produced from such a bundle: $y \leq f(x_1, \dots, x_n)$

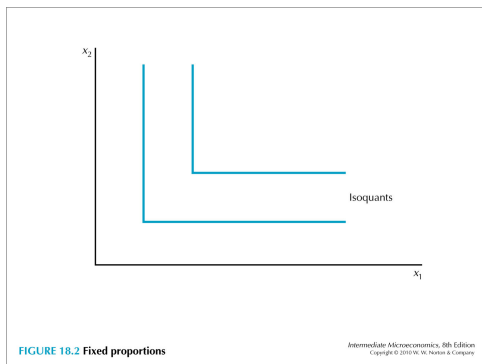
Isoquant of production

- ▶ A set of all possible combinations of inputs that are just sufficient to produce a given amount of output
- ▶ $Q(y^0) = \{(x_1, \dots, x_n) \in \mathbb{R}_{0,+}^n | f(x_1, \dots, x_n) = y^0\}$

EXAMPLES OF TECHNOLOGIES

Fixed proportions (Leontief technology)

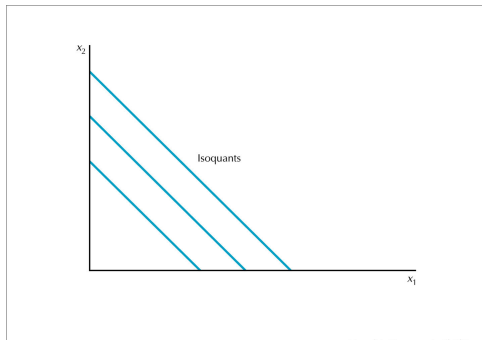
- ▶ One input cannot be replaced by the other
- ▶ $y = f(x_1, x_2) = \min\{ax_1, bx_2\}$
- ▶ To produce one unit of output we need $\alpha = \frac{1}{a}$ units of x_1 and $\beta = \frac{1}{b}$ units of x_2
- ▶ The slope of the line connecting the kink points is equal to a/b



EXAMPLES OF TECHNOLOGIES

Perfect substitutes

- ▶ Easy to switch between inputs.
- ▶ $y = f(x_1, x_2) = ax_1 + bx_2$
- ▶ To produce one unit of output we need $\alpha = \frac{1}{a}$ units of x_1 or $\beta = \frac{1}{b}$ units of x_2
- ▶ $a(b)$ tells us how many units of output we can produce from one unit of $x_1(x_2)$
- ▶ The slope of the isoquant is equal to $-a/b$



EXAMPLES OF TECHNOLOGIES

Cobb-Douglas production function

- ▶ Less than perfect substitutes, less than perfect complements.
- ▶ $y = f(x_1, x_2) = Ax_1^a x_2^b$
- ▶ A measures the scale of production.
- ▶ a and b measure responsiveness of output to changes in the inputs.

PROPERTIES OF TECHNOLOGY

Assumptions about production function and isoquants

- ▶ Inputs and outputs are non-negative
- ▶ Essentiality: $f(0, \dots, 0) = 0$
 - ▶ Strong essentiality: $\exists i : x_i = 0 \Rightarrow f(x_1, \dots, x_n) = 0$
- ▶ Inefficient production is possible:
 $\forall (x_1, \dots, x_n) : 0 \leq y \leq y_{max} = f(x_1, \dots, x_n)$
- ▶ Perfect divisibility of all inputs and outputs
- ▶ Continuity of the production function
- ▶ Smoothness of the production function
- ▶ Monotonicity of the production function: $\forall x \in X \exists i : \frac{\partial f}{\partial x_i} > 0$
- ▶ Convexity (isoquants bend towards the origin):
 $f(\lambda x_1 + (1 - \lambda)z_1, \lambda x_2 + (1 - \lambda)z_2) \geq f(x_1, x_2) = f(z_1, z_2)$

PROPERTIES OF TECHNOLOGY

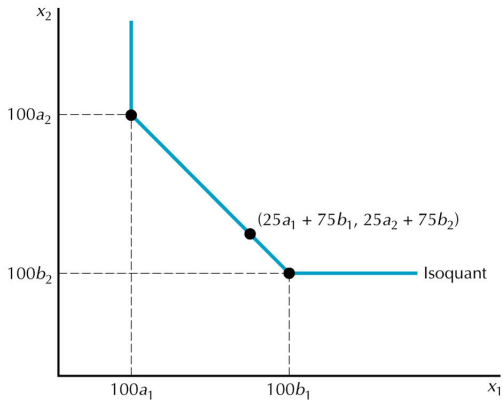


FIGURE 18.4 Convexity

Intermediate Microeconomics, 8th Edition
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MARGINAL PRODUCT

Marginal product of factor 1: Extra output from increasing the input of factor 1 a little bit and keeping factor 2 fixed.

$$\blacktriangleright MP_1(x_1, x_2) = \frac{\Delta y}{\Delta x_1} = \frac{f(x_1 + \Delta x_1, x_2) - f(x_1, x_2)}{\Delta x_1}$$

$$\blacktriangleright MP_i = \frac{\partial f}{\partial x_i}(x)$$

Assumptions

- ▶ Monotonic technology: MP non-negative
- ▶ Law of diminishing marginal product: $\frac{\partial MP_1(x_1, x_2)}{\partial x_1} \leq 0$

MARGINAL PRODUCT, APPLICATION

Microcredit - promise to alleviate poverty

- ▶ Loans to poor people, who were thought to be "unbankable" (no collateral, illiteracy).
- ▶ M. Yunus, Nobel peace prize, 74 million clients in 2009.
- ▶ Repayment rate above 95%.

Important assumption: high returns to investment for the poor.



MARGINAL PRODUCT, APPLICATION

Measuring returns to capital: deMel, McKenzie, Woodruff (2008)

- ▶ Sample: 400 microenterprises in Sri Lanka (small grocery stores, sewing clothing, making bamboo products, repairing bicycles, etc.)
- ▶ Randomized grants: \$0, \$100 or \$200.
- ▶ Follow-up data collected in the next 15 months.
- ▶ Return to capital 5% per month.

DESCRIPTION OF ISOQUANTS OF PRODUCTION

- ▶ Slope
- ▶ Curvature
- ▶ Density

TECHNICAL RATE OF SUBSTITUTION

Slope of the isoquant

$$\blacktriangleright \Delta y = MP_1(x_1, x_2) \Delta x_1 + MP_2(x_1, x_2) \Delta x_2 = 0$$

$$TRS(x_1, x_2) = \frac{\Delta x_2}{\Delta x_1} = -\frac{MP_1(x_1, x_2)}{MP_2(x_1, x_2)}$$

$$\blacktriangleright df(x) = \frac{\partial f}{\partial x_1}(x) \cdot dx_1 + \frac{\partial f}{\partial x_2}(x) \cdot dx_2 = 0$$

$$\frac{dx_2}{dx_1} = -\frac{\frac{\partial f}{\partial x_1}(x)}{\frac{\partial f}{\partial x_2}(x)} = -\frac{MP_1}{MP_2}$$

Assumptions

- ▶ Diminishing TRS (isoquants have convex shape)

TECHNICAL RATE OF SUBSTITUTION



Exercise

Production function $f(x) = \sqrt{x_1 + 1} + \sqrt{x_2 + 1} - 2$

Derive technical rate of substitution for $(x_1, x_2) = (15, 8)$

TECHNOLOGICAL EFFICIENCY OF PRODUCTION

Technological efficiency: Not possible to produce the output with smaller amount of at least one input while the amount of other inputs is held constant.

A cost-minimizing firm will always choose a point in the **economic region** , i.e. where the slope of the isoquant is negative

Output efficiency of production: $y = f(x)$

ELASTICITY OF SUBSTITUTION

Curvature of the isoquant can be described by the rate by which the ratio of inputs changes when we change the slope of the isoquant.

- ▶ High for substitutes
- ▶ Low for complements

$$\sigma = \frac{\% \Delta \frac{x_j}{x_i}}{\% \Delta |TRS_{ji}|} = \frac{\frac{\Delta \frac{x_j}{x_i}}{\frac{x_j}{x_i}}}{\frac{\Delta |TRS_{ji}|}{|TRS_{ji}|}}$$

$$\sigma = \frac{1}{\frac{d|TRS_{ji}|}{d(\frac{x_j}{x_i})}} \cdot \frac{|TRS_{ji}|}{\frac{x_j}{x_i}}$$

ELASTICITY OF SUBSTITUTION



Exercise

Production function $f(x) = 3\sqrt{x_1} + 4\sqrt{x_2}$

Derive the elasticity of substitution

ELASTICITY OF SUBSTITUTION

CES function (constant elasticity of substitution)

- ▶ $y = A(a_1x_1^\alpha + a_2x_2^\alpha)^{\frac{1}{\alpha}}, A > 0, a_1 + a_2 = 1$
- ▶ $TRS_{21} = -\frac{a_1x_1^{\alpha-1}}{a_2x_2^{\alpha-1}}$

Special cases

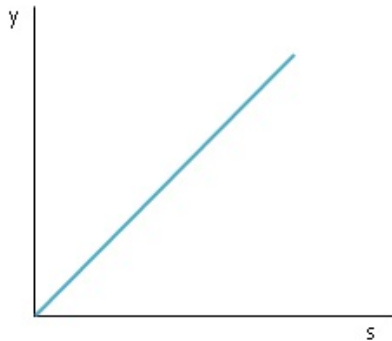
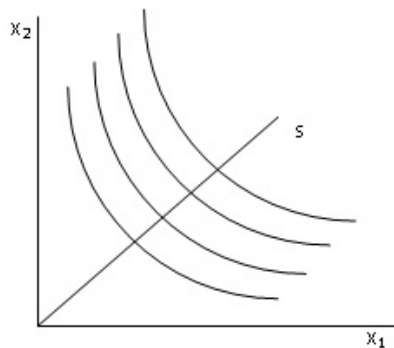
- ▶ Linear production function (perfect substitutes)
 - ▶ $\alpha = 1, \sigma \rightarrow \infty$
- ▶ Cobb-Douglas production function
 - ▶ $\alpha \rightarrow 0, \sigma = 1$
- ▶ Leontief production function (perfect complements)
 - ▶ $\alpha \rightarrow -\infty, \sigma \rightarrow 0$

RETURNS TO SCALE

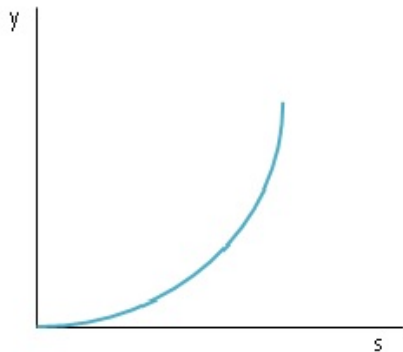
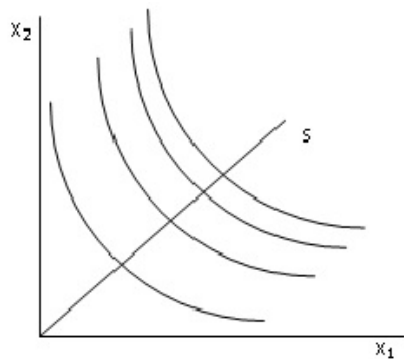
Change in output when we scale the amount of all inputs by constant factor

- ▶ Constant returns to scale
 - ▶ $f(sx_1, sx_2) = sf(x_1, x_2)$
 - ▶ $\frac{y}{s}$ is constant in s
 - ▶ Replicate what has been done before.
 - ▶ Consistent with diminishing *MP* to each factor.
- ▶ Increasing returns to scale
 - ▶ $f(sx_1, sx_2) > sf(x_1, x_2)$
 - ▶ $\frac{y}{s}$ is increasing in s
 - ▶ Not necessary to increase all inputs to increase output, synergies
- ▶ Decreasing returns to scale
 - ▶ Peculiar case
 - ▶ $f(sx_1, sx_2) < sf(x_1, x_2)$
 - ▶ $\frac{y}{s}$ is decreasing in s
 - ▶ Inputs not homogenous, loss of efficiency of organization

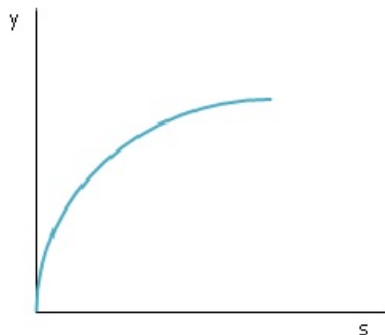
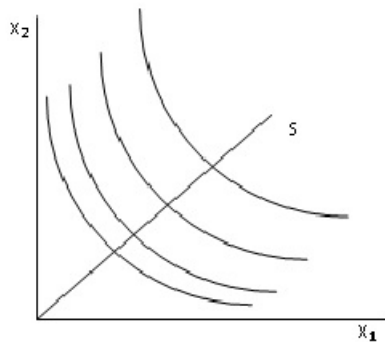
CONSTANT RETURNS TO SCALE



INCREASING RETURNS TO SCALE



DECREASING RETURNS TO SCALE



ELASTICITY OF RETURNS TO SCALE

The effect of the relative change in scale of production on the relative change in production.

$$e_s^y = \frac{\frac{dy}{y}}{\frac{ds}{s}} = \frac{dy}{ds} \frac{s}{y}$$

$$e_s^y = \frac{df(sx)}{ds} \frac{s}{f(sx)}$$

- ▶ Constant returns to scale: $e_s^y = 1$
- ▶ Increasing returns to scale: $e_s^y > 1$
- ▶ Decreasing returns to scale: $e_s^y < 1$

ELASTICITY OF RETURNS TO SCALE



Exercise

Production function $f(x) = 3\sqrt{x_1} + 4\sqrt{x_2}$

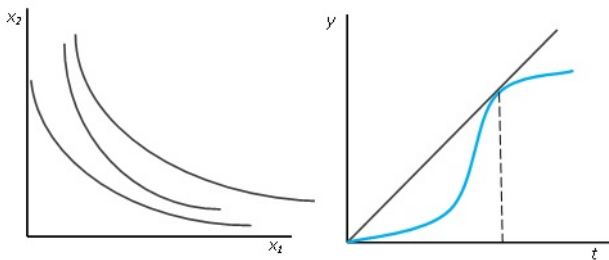
Derive the elasticity of returns to scale. Are they constant, increasing or decreasing?

ELASTICITY OF RETURNS TO SCALE

Technology can exhibit different kind of returns to scale for different ratios of inputs.

Technology can exhibit different kind of returns to scale at different levels of production.

- Minimum efficient scale of production: highest $\frac{y}{s}$



HOMOGENOUS PRODUCTION FUNCTION

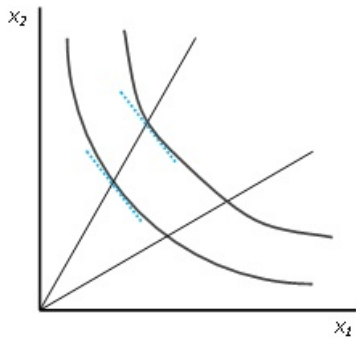
Production function: $f(sx) = s^t f(x)$

Linearly homogenous production function: $t = 1$

Useful properties

- ▶ Elasticity of returns to scale is constant and equal to t .
- ▶ Function homogenous of degree t : MP homogenous of degree $t - 1$.
- ▶ TRS depends on the ratio of inputs, not on the scale of production.

HOMOGENOUS PRODUCTION FUNCTION



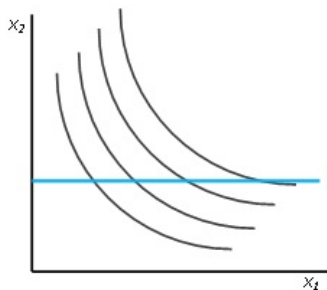
SHORT-RUN

Short-run: some factors of production are fixed

Long-run: all factors of production can be varied

No specific time interval

Short-run production function: $f(x_1, \bar{x}_2)$



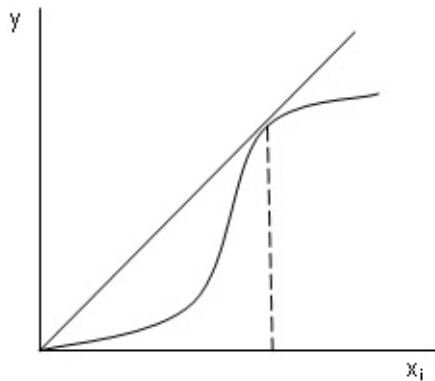
SHORT-RUN

Average product: $AP_i(x) = \frac{y}{x_i} = \frac{f}{x_i}(x)$

Returns to factor

- ▶ Elasticity of returns to factor i: $e_{x_i}^y = \frac{MP_i}{AP_i} = \frac{\frac{\partial f(x_i)}{\partial x_i}}{\frac{f(x_i)}{x_i}}$
- ▶ Increasing returns: AP increasing, $MP > AP$, $e_{x_i}^y > 1$
- ▶ Decreasing returns: AP decreasing, $MP < AP$, $e_{x_i}^y < 1$
- ▶ $MP = AP$, $e_{x_i}^y = 1$: constant returns to factor, or change from increasing to decreasing or vice versa

SHORT-RUN



SHORT-RUN



Exercise

Production function $f(x) = (\sqrt{x_1} + \sqrt{x_2})^3$ and in the short-run $x_2 = 4$

Derive the short-run production function

Calculate the elasticity of returns to factor 1 and find the amount of x_1 where the returns change from increasing to decreasing

RELATIONSHIP BETWEEN MP AND AP

